

Descriptive Statistics for Bivariate Regression (Part 1)

August 30, 2022

Outline

Univariate Descriptive Statistics

Descriptive Regression with a Coarse Covariate

Descriptive Linear Regression

What are descriptive statistics?

- **Descriptive statistics are summaries of the data we have.**
- The data we have might be a sample (e.g., an exit poll) or it might be the whole population (e.g., administrative vote counts), sometimes known as a census.
- Having the whole population is becoming more common in the era of Big Data.
- Note: if you are interested in the causal effects of a treatment (a drug, a policy, etc.), you can never really have the whole population (we will discuss this in the causal inference portion of the course).

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What makes a good summary?

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From the google definition:

1. brief
2. includes main points

What makes a good descriptive statistic?

1. brief: at most a few numbers

2. includes main points:

- sufficient to describe main characteristics of the data
 - there is a technical definition of sufficiency that I can discuss in office hours for those interested
- minimizes the information lost by reducing the data
 - I'm using "information" in a loose sense here, will formalize a bit later
 - reducing the data means taking the data, which are often represented by $n * v$ numbers (with n the number of observations and v the number of variables) and reducing to a descriptive statistic that is represented by a few numbers.

Don't worry. Enough forest, trees (examples) are coming!

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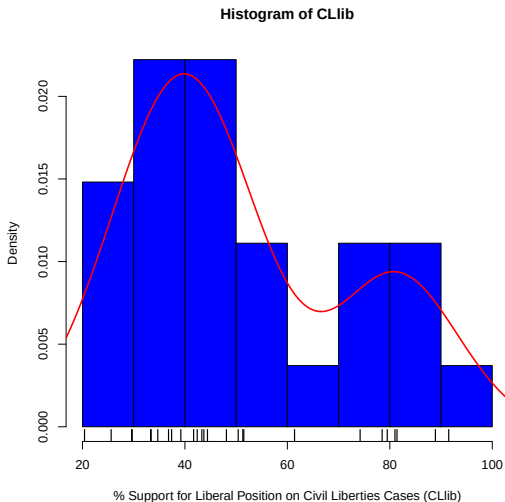
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"Hold on, where's the forest again?"

It is easy in a statistics class to focus on the specifics. I want you to remember why we are doing what we are doing.

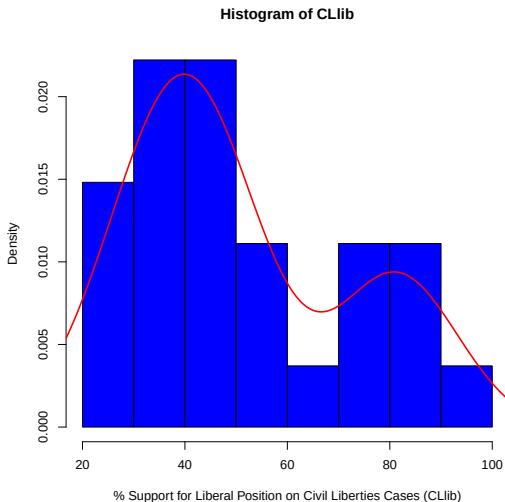
Describing Univariate Data



Summarization

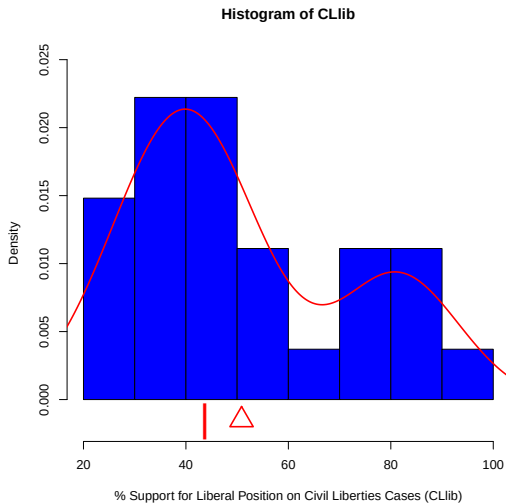
- Even for univariate data, we may want to focus on one or two characteristics of the population instead of describing it in its entirety.
- We typically focus on measures of center and spread.
- We will discuss the rationale for this over the coming weeks.

Univariate Activity

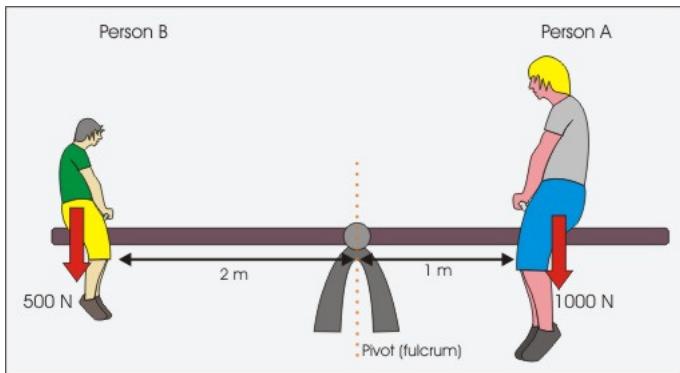


Find the rough location of the mean and median.

Univariate Activity Solution



Mean as a balance point



Choosing a descriptive statistic

How might we choose between these descriptive statistics?

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A Least Squares Summary

Suppose we want to pick a single number (m) for the middle of the data that summarizes all the values for y , by minimizing the sum of squared residuals (i.e., least squares).

$$SSR(\tilde{m}) = \sum_{i=1}^N (y_i - \tilde{m})^2.$$

Activity: Using calculus, derive the least squares estimator

1. Calculate the derivative of SSR with respect to $\tilde{m}$
2. Set the derivative equal to 0
3. Solve for m

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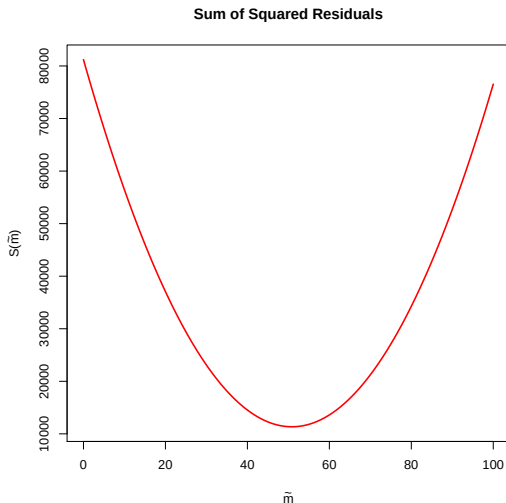
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The Objective Function for SSR CLlib

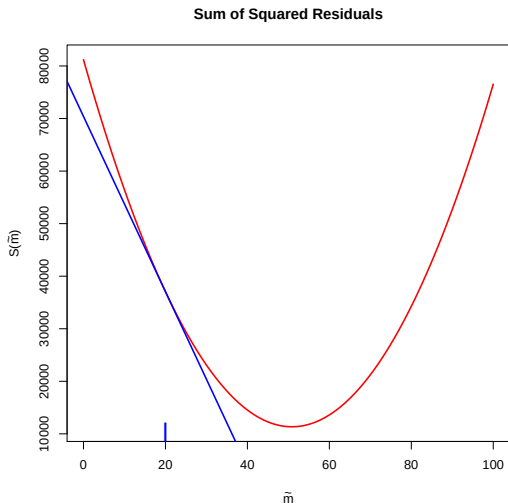


1.

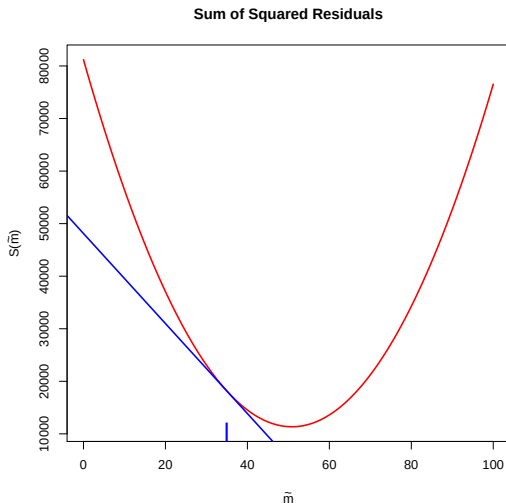
$$\begin{aligned}
 SSR(\tilde{m}) &= \sum_{i=1}^N (y_i - \tilde{m})^2 \\
 &= \sum_{i=1}^N (y_i^2 - 2y_i\tilde{m} + \tilde{m}^2)
 \end{aligned}$$

$$\frac{\partial SSR(\tilde{m})}{\partial \tilde{m}} = \sum_{i=1}^N (-2y_i + 2\tilde{m})$$

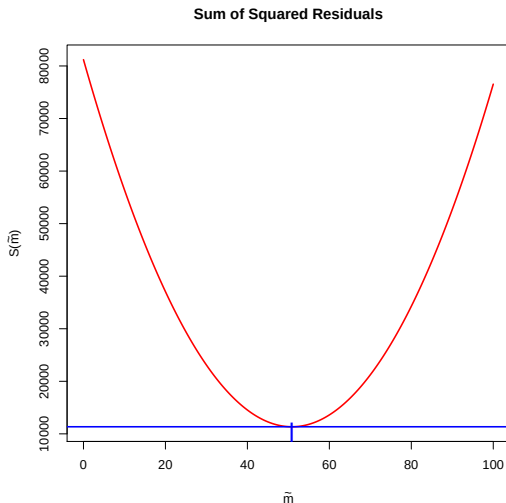
The Slope of the Tangent Line for SSR CLlib



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$$\frac{\partial S(m)}{\partial \tilde{m}} = \sum_{i=1}^N (-2y_i + 2\tilde{m})$$

2.

$$0 = \sum_{i=1}^N (-2y_i + 2m)$$

3.

$$m = \frac{1}{N} \sum_{i=1}^N y_i$$

Therefore, the mean is a least squares summary.

Why we focus on means

- Means often provide a “good” summary of the data (and it is relatively easy to tell when they provide bad summaries).
- The properties of means are relatively easy to describe.
- Sufficient in a sense (in a few slides).
- Randomized experiments only identify average causal effects (as do regression and matching when the appropriate assumptions hold).

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Variance and Standard Deviation

- Mean is the least squares predictor in terms of SSR.
- How small is “least” (i.e., how much spread around the mean)?
- Variance and Standard Deviation are useful transformations of SSR for the mean.

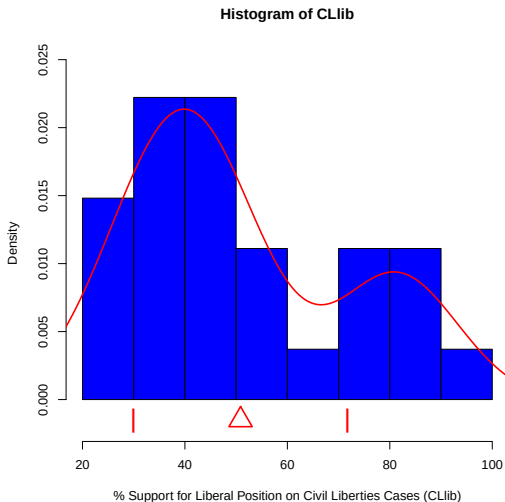
Population Variance

$$\begin{aligned} S^2 &= \frac{\sum_{i=1}^N (y_i - \bar{y})^2}{N - 1} \\ &= \frac{SSR}{N - 1} \end{aligned}$$

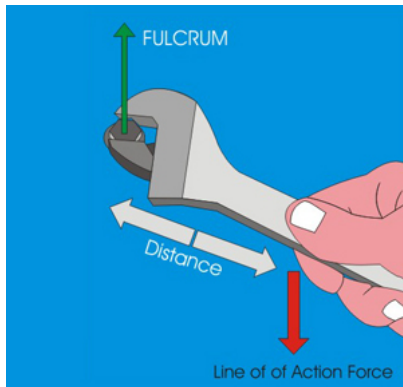
Population Standard Deviation

$$S = \sqrt{S^2}$$

Mean and Standard Deviation



Variance as turning force



Why variance and standard deviation?

- Natural measures of accuracy for means.
- Means and variances are sufficient for normally distributed data.
- For nonnormal data, means and variances provide much information via Chebyshev's inequality and other concentration inequalities.

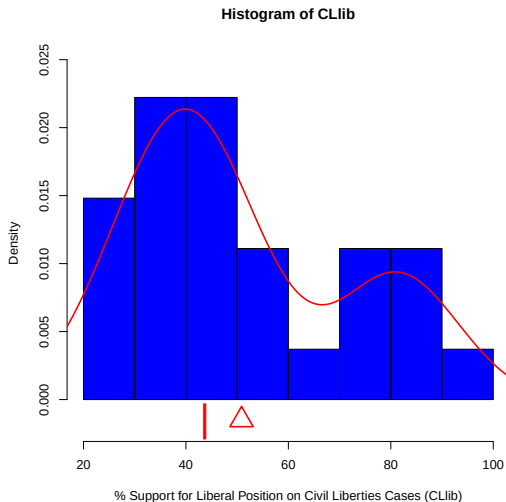
Binary Outcome with Binary Covariate

Table: Population

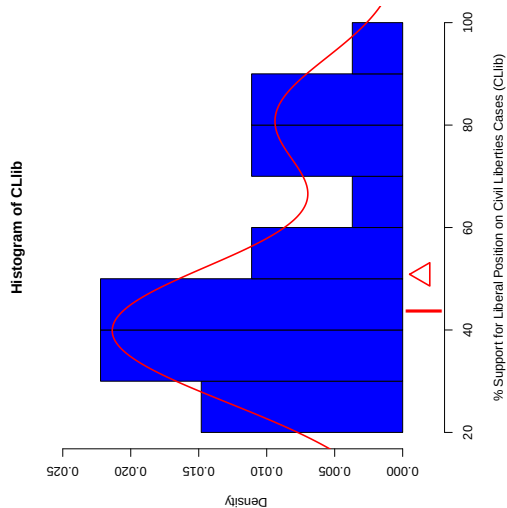
i	1	2	3	4	...	625	mean
y_i	1	1	0	1	...	0	.68
x_i	1	0	1	0	...	0	.5

- Instead of one proportion, we now have two conditional proportions (one for $x = 0$ and one for $x = 1$).
- In regression, we focus on conditional means/proportions (not joint proportions).

From Univariate to Regression



From Univariate to Regression



Coarse and Fine-grained Covariates

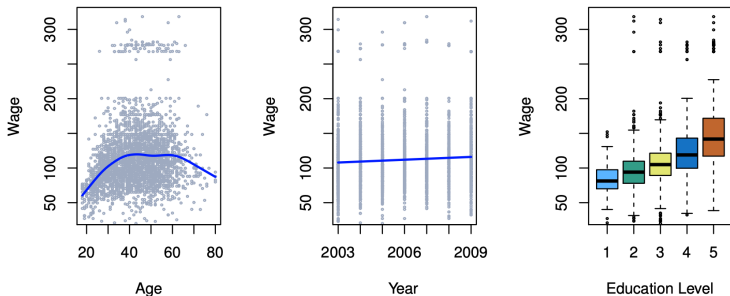
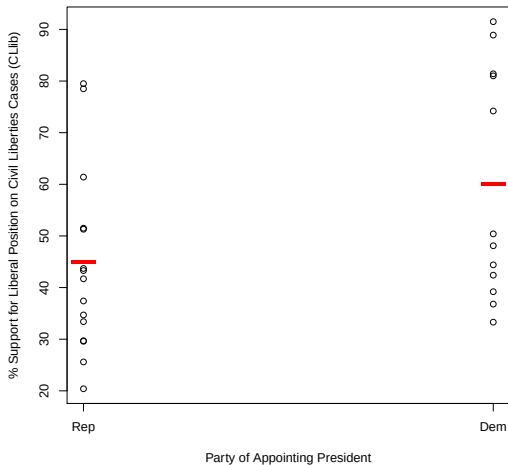
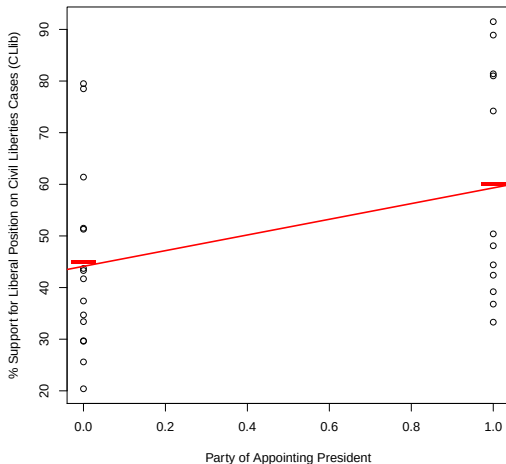


FIGURE 1.1. Wage data, which contains income survey information for men from the central Atlantic region of the United States. Left: **wage** as a function of **age**. On average, **wage** increases with **age** until about 60 years of age, at which point it begins to decline. Center: **wage** as a function of **year**. There is a slow but steady increase of approximately \$10,000 in the average **wage** between 2003 and 2009. Right: Boxplots displaying **wage** as a function of **education**, with 1 indicating the lowest level (no high school diploma) and 5 the highest level (an advanced graduate degree). On average, **wage** increases with the level of education.

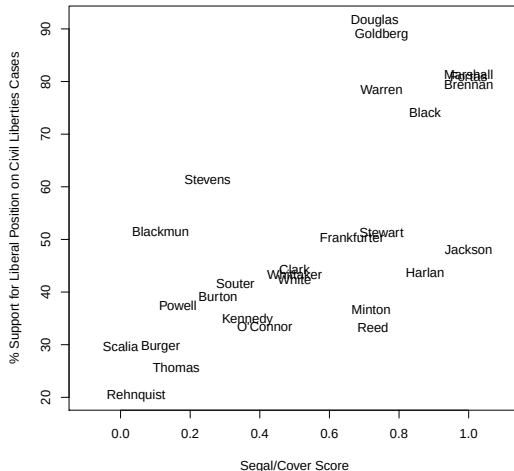
Mean CLib Conditional on Party



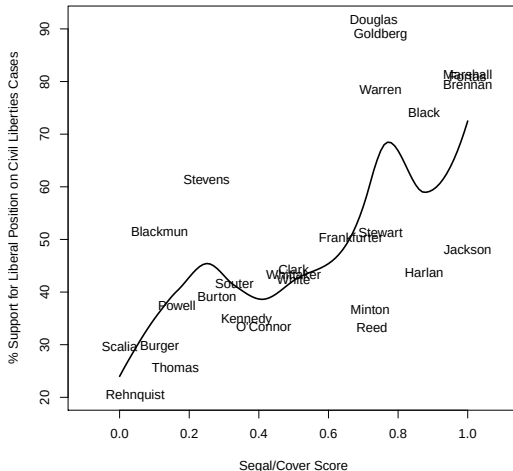
Linear Conditional Mean Function (LCMF)



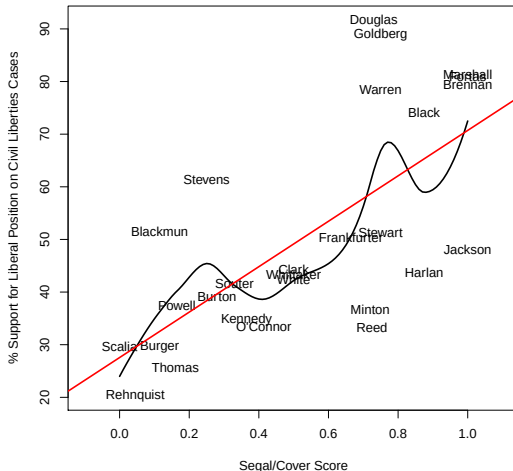
What about a continuous explanatory variable?



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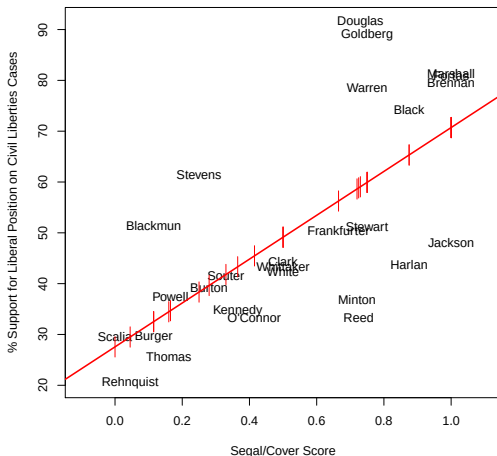


What about a continuous explanatory variable?



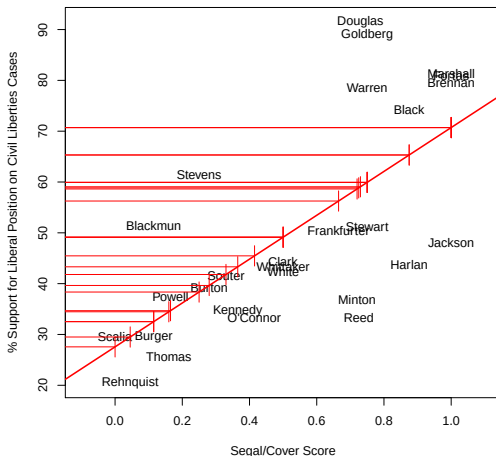
Fitted Values and Residuals

- Fitted Values from LCMF: $\hat{y}_i = B_0 + B_1 x_i$ for $i = 1, \dots, N$
- Residuals from Fitted Values: $e_i = y_i - \hat{y}_i$ for $i = 1, \dots, N$



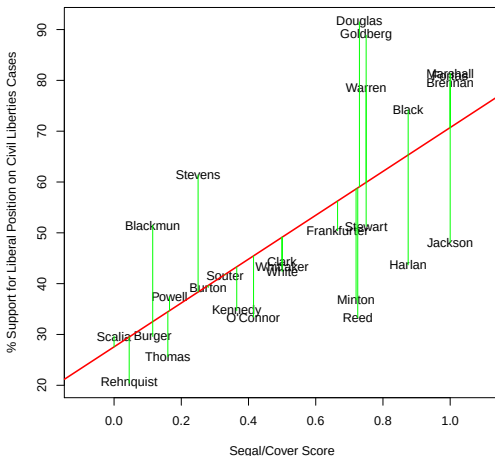
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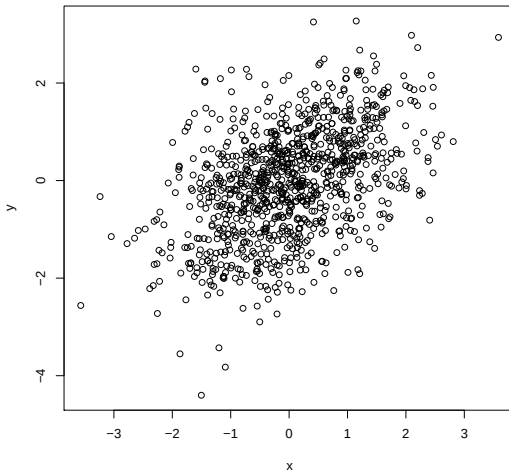


Minimizing the Residuals

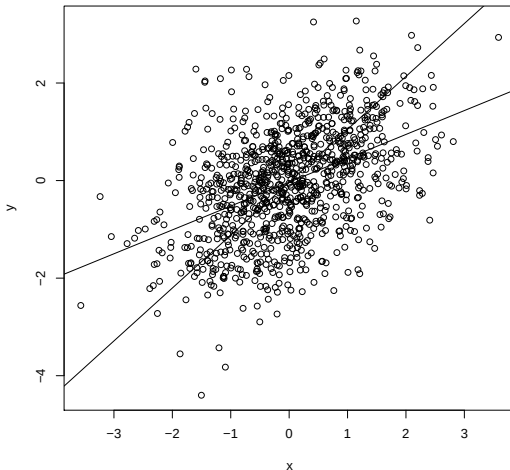
- Let $\tilde{B}_0$ and $\tilde{B}_1$ be possible values for B_0 and B_1 respectively,
- *Least Squares (LS) regression:*

$$(B_0, B_1) = \operatorname{argmin}_{\tilde{B}_0, \tilde{B}_1} \sum_{i=1}^N (y_i - \tilde{B}_0 - x_i \tilde{B}_1)^2$$

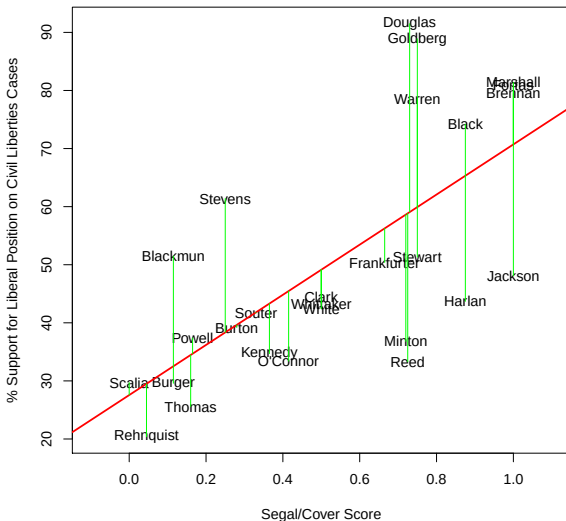
Simple Regression Activity



Simple Regression Activity



Choosing Slope and Intercept to Minimize SSR



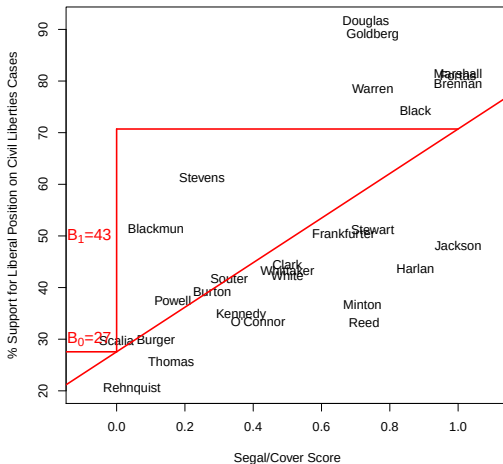
Deriving the Linear Least Squares Estimator

Let $\tilde{B}_0$ and $\tilde{B}_1$ be possible values for B_0 and B_1 respectively, and

$$S(\tilde{B}_0, \tilde{B}_1) = \sum_{i=1}^N (y_i - \tilde{B}_0 - x_i \tilde{B}_1)^2.$$

1. Take partial derivatives of S with respect to $\tilde{B}_0$ and $\tilde{B}_1$.
2. Set each of the partial derivatives to 0
3. Substitute B_0 and B_1 for $\tilde{B}_0$ and $\tilde{B}_1$ and solve for B_0 and B_1

Descriptive Interpretation



Variance and Regression

- Again, we want to assess the “leastness” of least squares (i.e., the spread of the data around the line).
- The standard error of the regression, $\sqrt{\frac{\sum_{i=1}^N e_i^2}{N-2}}$, is analogous to the standard deviation.
- There are other approaches to this question: correlation and coefficient of determination.

Covariance

The population variances for X and Y can be written as:

$$\begin{aligned} S_x^2 &= \frac{\sum_{i=1}^N (x_i - \bar{x})^2}{N - 1} \\ &= \frac{\sum_{i=1}^N (x_i - \bar{x})(x_i - \bar{x})}{N - 1} \end{aligned}$$

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This quantity is the **covariance**.

Correlation

The covariance does not have units that are easy to interpret, so we can scale it by the standard deviations of X and Y to obtain the **correlation**:

$$R = \frac{S_{xy}}{S_x S_y} = \frac{\sum_{i=1}^N (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^N (x_i - \bar{x})^2 \sum_{i=1}^N (y_i - \bar{y})^2}}$$

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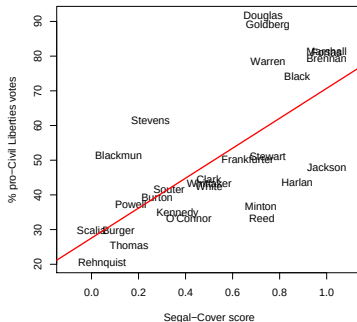
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Analysis of Variance for the Regression

How well do the Segal-Cover scores predict support for civil liberties in relation to the average support for civil liberties?

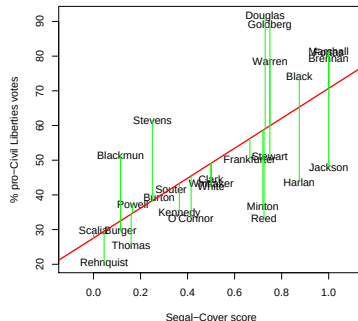
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- $(y_i - \bar{y}) = e_i + (\hat{y}_i - \bar{y})$



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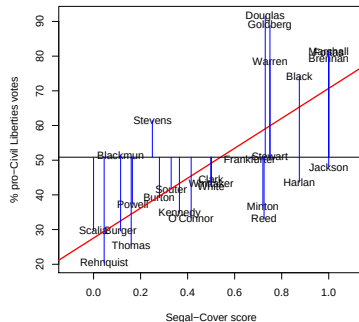
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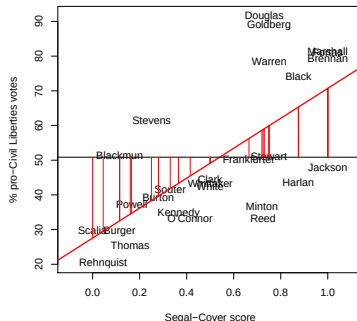
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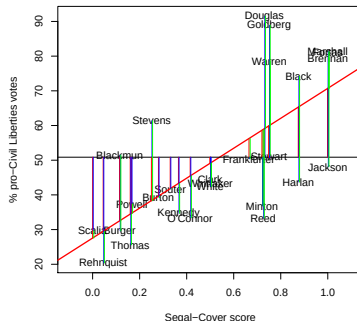
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- $(y_i - \bar{y}) = e_i + (\hat{y}_i - \bar{y})$



$$\sum_{i=1}^N (y_i - \bar{y})^2 = \sum_{i=1}^N \{e_i + (\hat{y}_i - \bar{y})\}^2$$

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Coefficient of Determination

We can divide each side by the TSS:

$$\frac{SSR}{TSS} + \frac{RegSS}{TSS} = \frac{TSS}{TSS}$$

$$\frac{SSR}{TSS} + \frac{RegSS}{TSS} = 1$$

$$\frac{RegSS}{TSS} = 1 - \frac{SSR}{TSS} = R^2$$

R^2 is a measure of how much of the variation in Y is accounted for by X .

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Usefull Identities

$$R^2 = (R)^2$$

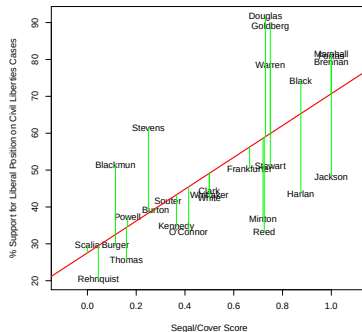
$$B_1 = \frac{\sum_{i=1}^N (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^N (x_i - \bar{x})^2}$$

Usefull Identities

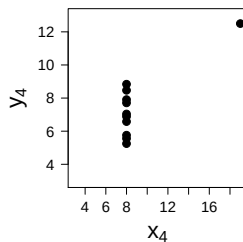
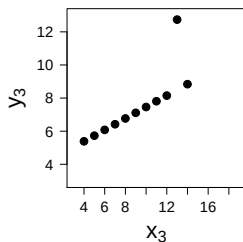
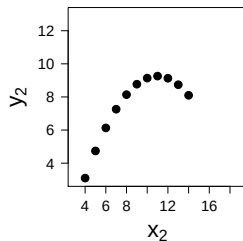
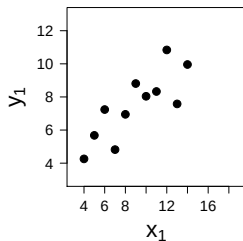
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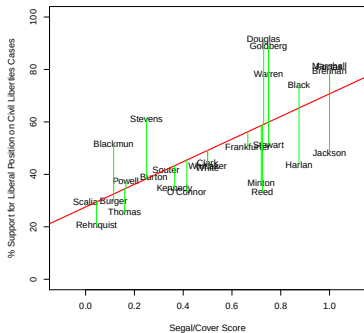
- $R^2 = 0.45$
- $R = \sqrt{R^2} = 0.67$
- $S_x = 0.33$
- $S_y = 20.9$
- $B_1 = R \frac{S_y}{S_x} = 43.15$



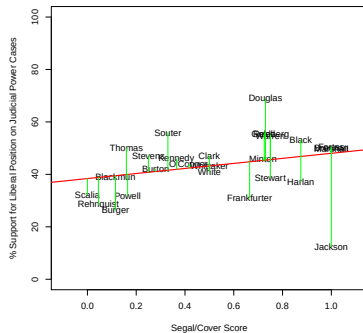
Anscombe's Quartet



Civil Liberties ($R^2 = 45\%$)



Judicial Power ($R^2 = 7\%$)



Tuesday Summary

- We will focus on conditional means as a good conditional summary of a single outcome variable.
- Conditional means are straightforward for a coarse covariate.
- Linear regression provides a parsimonious summary for a fine-grained covariate.

On Thursday

- Binary outcomes with a fine-grained covariate
- Flexible (i.e., ML) approaches